

CS240 Algorithm Design and Analysis
Spring 2026
Problem Set 2

Due: 23:59, April 12, 2026

1. Please write your solutions in English.
2. Submit your solutions to **Gradescope**. When submitting, match your solutions to the problems correctly.
3. If you want to submit a handwritten version, scan it clearly.
4. Up to 2 late days are allowed for homework submissions. Assignments submitted within the first 24 hours after the deadline will incur a 10% penalty; those submitted within 48 hours will incur a 20% penalty. Submissions more than 48 hours late will not be accepted unless there is a valid reason, such as a medical or family emergency.
5. You are required to follow ShanghaiTech's academic honesty policies. You are allowed to discuss problems with other students, but you must write up your solutions by yourselves. You are not allowed to copy materials from other students or from online or published resources. Violating academic honesty can result in serious penalties.

Problem 1:

Given a sequence of integers $a_1, a_2, \dots, a_N$, an increasing subsequence is a subsequence $a_{i_1}, a_{i_2}, \dots, a_{i_k}$, such that $1 \leq i_1 < i_2 < \dots < i_k \leq N$ and $a_{i_1} < a_{i_2} < \dots < a_{i_k}$. Your task is to find the maximum possible sum among all increasing subsequences of the given sequence.

Note that the longest increasing subsequence does not necessarily have the largest sum. Provide the state transition equation and the corresponding pseudocode.

Example 1:

Input: $a = [1, 7, 3, 5, 9, 4, 8]$

Output: 18

Explanation: One optimal increasing subsequence is $(1, 3, 5, 9)$, whose sum is 18.

Example 2:

Input: $a = [100, 1, 2, 3]$

Output: 100

Explanation: Although $(1, 2, 3)$ is longer, its sum is only 6. The subsequence (100) has the maximum sum.

Solution:

Problem 2:

A student plans to select courses for the coming semester. There are m course groups, and each group contains several candidate courses. We denote a course by $X(c, v)$, where X is the course name, c denotes its credit cost, and v denotes its benefit value.

The student may choose at most one course from each group, and the total number of credits used cannot exceed W . Your task is to determine the maximum total benefit that can be obtained under the credit limit. Provide the dynamic programming formulation and the corresponding pseudocode.

Example 1:

Input: $m = 3$, $W = 5$

Group 1: $A(1, 2)$, $B(3, 4)$

Group 2: $C(2, 2)$, $D(4, 5)$

Group 3: $E(1, 1)$, $F(2, 3)$, $G(3, 4)$

Output: 7

Explanation: An optimal selection is to choose course $A(1, 2)$ from Group 1, course $C(2, 2)$ from Group 2, and course $F(2, 3)$ from Group 3. The total credit cost is $1 + 2 + 2 = 5$, and the total benefit is $2 + 2 + 3 = 7$.

Example 2:

Input: $m = 2$, $W = 4$

Group 1: $A(1, 2)$, $B(2, 3)$

Group 2: $C(2, 4)$, $D(3, 5)$

Output: 7

Explanation: An optimal selection is to choose course $B(2, 3)$ from Group 1 and course $C(2, 4)$ from Group 2. The total credit cost is $2 + 2 = 4$, and the total benefit is $3 + 4 = 7$.

Solution:

Problem 3:

On a wall composed of a line of grid cells, there is a painting robot. The robot follows these two painting rules:

1. The robot can only paint using one color at a time, and it can only paint a continuous area of the same color.
2. The robot can start painting from any grid cell and can overwrite areas that have been previously painted.

Given a string s , where each character represents the color of a grid cell (e.g., “r” for red, “b” for blue, “g” for green), your task is to calculate the minimum number of painting operations needed for the robot to cover all the grid cells. Provide the state transition equation and the corresponding pseudocode.

Example 1:

Input: $s = \text{“rrrbbb”}$ Output: 2

Explanation: First, paint the red area “rrr”, then paint the blue area “bbb”.

Example 2:

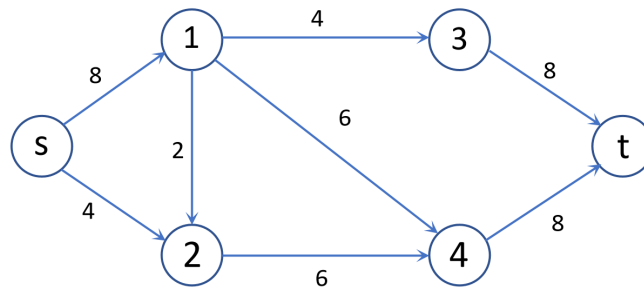
Input: $s = \text{“rbr”}$ Output: 2

Explanation: First, paint the entire row with red “rrr”, then overwrite the middle cell with blue “b”.

Solution:

Problem 4:

Execute the Ford-Fulkerson algorithm on the flow network shown in the figure below. Display the residual network after each flow update and determine the maximum flow at the end. During each iteration, choose the lexicographically smallest available augmenting path. For example, if two augmenting paths are $s \rightarrow 2 \rightarrow t$ and $s \rightarrow 3 \rightarrow t$, select $s \rightarrow 2 \rightarrow t$ since it is lexicographically smaller. Likewise, if the paths $s \rightarrow 3 \rightarrow t$ and $s \rightarrow 2 \rightarrow 4 \rightarrow t$ are available, choose $s \rightarrow 2 \rightarrow 4 \rightarrow t$.



Solution:

Problem 5:

Before the beginning of a new semester at ShanghaiTech University, there are n students who need to register for courses and m available courses. The registration process is subject to the following constraints:

1. Eligibility constraint: Each student may only register for courses that are allowed under their major requirements.
2. Student course limit: Each student i may enroll in at most k_i courses during the semester.
3. Course capacity limit: Each course j has a capacity limit and can admit at most c_j students.

Your task is to formulate this problem as a network flow model and explain the maximum number of valid student-course assignments subject to the constraints above.

Solution:

Problem 6:

Yansen, a student at ShanghaiTech, recently started playing Animal Crossing. The game follows real-time (a week has 7 days).

There are n events, each requiring different materials, and each event is available on a few days of the week. Yansen can complete up to e events per day due to academic and research commitments.

Yansen wants to build a special Gundam and has calculated the materials needed. Starting on Monday, what is the minimum number of days (including rest days) to collect all the materials? (Hint: Network Flow)

Input Description

The first line contains two integers, n, e ($1 \leq n \leq 1000, 1 \leq e \leq 100$), separated by a space, representing the number of different events and the maximum number of events Yansen can complete in a day.

The next n lines, each containing $m_i + 2$ integers $c_i, m_i, a_{i1}, a_{i2}, \dots, a_{im_i}$ ($1 \leq c_i \leq 10^5, 1 \leq m_i \leq 7, 1 \leq a_{ij} \leq 7$, for any $j \neq k, a_{ij} \neq a_{ik}$), represent the required number of completions c_i , and the days of the week when the event is available.

Output Description

Output a single integer representing the minimum number of days Yansen needs to gather all the required materials.

Input:

```
5 2
2 4 1 3 5 7
2 3 2 4 6
2 2 1 2
2 2 3 4
2 2 5 6
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Output: 5

Explanation:

The solution can proceed as follows:

Day 1 (Monday): Complete events 1 and 3 once each.

Day 2 (Tuesday): Complete events 2 and 3 once each.

Day 3 (Wednesday): Complete events 1 and 4 once each.

Day 4 (Thursday): Complete events 2 and 4 once each.

Day 5 (Friday): Complete event 5 twice.

Exactly 5 days are needed to finish all requirements.

Solution: